

Adversarial Variational Information Bottleneck for Semantic Image Watermarking: Overcoming Decoder Bypass and Collusion Estimation via SkipDropout and Chaotic Pooling

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Abstract—Deep convolutional autoencoders deployed for digital watermarking predominantly utilize unconstrained residual skip connections to preserve carrier edge fidelity. However, this architectural design introduces a critical systemic failure: the “lazy decoder” pathology, wherein the reconstruction network reconstructs carriers exclusively via skip shortcuts, effectively bypassing the latent bottleneck and treating watermark perturbations as disposable noise. Furthermore, the common practice of embedding a static spread-spectrum sequence across carriers leaves media forensic frameworks vulnerable to multi-image averaging and differential collusion estimation attacks. This paper proposes a robust, end-to-end deep generative watermarking architecture centered on an Adversarial Variational Autoencoder (VAE) equipped with stochastic SkipDropout and an anti-collusion chaotic watermark pool. Our framework decouples semantic foreground objects from background contexts, mapping representations into a continuous latent space regularized via Kullback–Leibler divergence. To eliminate the bypass shortcut, SkipDropout stochastically severs skip pathways during training ($p = 0.3$), forcing the decoder to reconstruct carriers directly from the watermarked latent code. Concurrently, dynamic watermarks are drawn from a 256-instance pool driven by SHA-512 cryptographic hashing and coupled Logistic-Sine chaotic maps, modulated into the latent domain using a fully differentiable, texture-adaptive semantic mask and dynamic α -scheduling. Distorted carriers undergo in-loop generative restoration before dual verification: payload bit-stream extraction and zero-shot adversarial authenticity discrimination. Evaluated over a 31,575-image corpus and cross-validated on Kodak and BOSSbase, our framework achieves an imperceptibility of 45.24 dB PSNR and 0.9967 SSIM, while sustaining normalized correlation values $NC \geq 0.91$ and bit error rates $BER \leq 0.06$ under severe JPEG compression ($QF = 50$), Gaussian noise, and cropping attacks, eliminating collusion estimation leakage.

Index Terms—Digital Image Watermarking, Variational Autoencoder, SkipDropout, Chaotic Maps, Media Forensics, Generative Adversarial Networks.

I. INTRODUCTION

The ubiquity of open multimedia distribution networks has escalated the requirement for verifiable media provenance, tamper detection, and intellectual property protection. While visible watermarks degrade perceptual utility and metadata

headers remain susceptible to stripping, imperceptible digital image watermarking embeds auxiliary cryptographic signatures directly into the host pixel distribution. An optimal watermarking framework must navigate an intrinsic three-way trade-off between perceptual transparency, extraction robustness, and data payload capacity. The carrier must remain visually indistinguishable from the pristine original, withstand aggressive signal processing and malicious geometric transformations, and securely retain sufficient payload bits to verify provenance. Early watermarking paradigms operated in deterministic transform domains, embedding spread-spectrum sequences into significant Discrete Cosine Transform (DCT) or Discrete Wavelet Transform (DWT) coefficients. Although computationally efficient, transform methods lack contextual adaptability, exhibiting severe vulnerability to non-linear distortions such as deep generative post-processing, adversarial noise, and aggressive lossy compression. Consequently, research has shifted toward end-to-end deep learning architectures, notably Convolutional Cascaded Autoencoders (CCAEs). CCAEs project cover images into low-dimensional latent bottlenecks, introduce watermark perturbations, and synthesize watermarked carriers via decoder subnets. To prevent high-frequency blurriness, these autoencoders incorporate dense residual skip connections between encoder and decoder layers.

Empirical investigations into deterministic skip-connected autoencoders reveal a critical structural vulnerability: the “lazy decoder” pathology. Because high-frequency spatial features flow unimpeded through skip connections, the decoder tends to reconstruct the carrier image almost exclusively from these bypass features, effectively ignoring the latent bottleneck. Consequently, the injected watermark perturbation is treated as disposable noise and smoothed out during the reconstruction phase. This phenomenon drives the gradient of the reconstruction loss with respect to the latent perturbation toward zero ($\partial \mathcal{L}_{\text{rec}} / \partial Z' \rightarrow 0$), resulting in watermark collapse when subjected to minor real-world distribution shifts.

Simultaneously, existing deep watermarking models suffer

from significant cryptographic and statistical weaknesses. The widespread practice of embedding a single, static watermark sequence across an entire dataset facilitates trivial estimation attacks. Adversaries can isolate the static sequence by averaging multiple watermarked carriers, thereby canceling out independent image content. Furthermore, attempts to constrain watermark visibility often rely on non-differentiable binary edge masks, which shatter gradients during backpropagation and preclude the end-to-end optimization of localized embedding strengths.

To address these interconnected failure modes, this paper introduces an end-to-end Adversarial Variational Information Bottleneck (AVIB) framework designed for robust semantic image watermarking. The primary contributions of this work are as follows:

- We identify and mathematically formalize the “lazy decoder” pathology in skip-connected watermarking autoencoders, demonstrating empirically that unconstrained skip pathways lead to latent bypass and watermark erasure.
- We propose a stochastic SkipDropout mechanism ($p = 0.3$) integrated within a ResNet-based Variational Autoencoder (VAE). This dynamically severs skip shortcuts during training, forcing bottleneck dependency and ensuring robust gradient flow through the watermark channel.
- We develop an anti-collusion chaotic watermark pool comprising 256 instances, driven by SHA-512 key expansion and coupled Logistic-Sine maps. This architecture prevents statistical estimation attacks while regularizing the forensic extraction process.
- We introduce a continuous, fully differentiable Adaptive Semantic Masker paired with epoch-dependent dynamic α -scheduling to maintain high imperceptibility (45.24 dB PSNR) without inducing gradient shattering.
- We implement an in-loop generative attack-inversion network and a dual-verification mechanism (forensic bit retrieval and adversarial authenticity discrimination) that sustains high robustness ($NC \geq 0.91$) against severe distortions down to JPEG $QF = 50$.

II. LITERATURE REVIEW, SCOPE AND CONTRIBUTIONS

Transform-domain watermarking historically represented the standard for copyright protection. Cox et al. [3] introduced spread-spectrum watermarking, injecting independent and identically distributed (i.i.d.) Gaussian sequences into the perceptually dominant DCT coefficients of an image. Subsequent schemes integrated multiple transforms; Begum et al. [2] concatenated DCT, DWT, and Singular Value Decomposition (SVD) to embed signatures within diagonal sub-bands, resisting geometric attacks. Al-Aiash et al. [2] leveraged Schur matrix factorization combined with hybrid DCT-DWT decompositions. While mathematically elegant, these handcrafted schemes rely on rigid bases and fail under non-linear perturbations, lossy compression, and neural resampling.

Deep convolutional autoencoders automated feature engineering by treating watermarking as an end-to-end optimization

problem. Zhu et al. pioneered this paradigm with HiDDeN, jointly training an encoder, forward noise layers, and a decoder subnet. Ahmaderaghi et al. [2] utilized discrete shearlet transforms and decision theory to guide blind extraction, while Loan et al. [2] explored chaotic encryption with coefficient differencing. Patra and Maity [3] designed a Cascaded Convolutional Autoencoder (CCAE) using differential latent dimensions for foreground and background regions, reaching 45.24 dB PSNR on Edge-IoT nodes. Patra et al. [2] further benchmarked spatial-statistical fusions, demonstrating that Spatial Kullback–Leibler Divergence (KLD) achieved 94% classification accuracy. Nevertheless, these deterministic models retain unconstrained skip connections, rendering the latent code susceptible to decoder bypass and information erasure. Goodfellow et al. [1] established the min-max adversarial framework, which has been widely adopted in media steganography and watermarking. Duan et al. [3] utilised improved Wasserstein GANs (WGAN-GP) for coverless steganography, synthesising carriers directly from hidden payloads. In active watermarking, Zhang et al. [3] introduced Dual Defence against face-swapping deepfakes, while Wu et al. [3] proposed SepMark, decoupling robust and fragile decoders. However, existing GAN detectors often suffer from training instability, requiring continuous in-loop generative refinement to condition degraded inputs prior to extraction. Chaotic sequences provide high sensitivity to initial states, non-periodic trajectories, and ergodicity, making them suitable for watermarking security. Classical 1D systems such as the Logistic map exhibit narrow chaotic parameter windows and non-uniform distributions. Coupling the Logistic and Sine maps eliminates periodic blank windows and expands the key space. However, repeatedly embedding a static chaotic sequence across a collection of images allows attackers to isolate the watermark via image averaging. Dynamic pooling is required to maintain anti-collusion security **Scope and Contribution**

III. MATHEMATICAL MODELS AND MEASURES

A. The “Lazy Decoder” Pathology

Let an input image $I \in \mathbb{R}^{3 \times H \times W}$ be compressed by an encoder $\mathcal{E}_\phi$ into a latent bottleneck $Z = \mathcal{E}_\phi(I)$, and let intermediate feature representations at layer $l \in \{1, \dots, L\}$ be forwarded via skip connections $\mathcal{S} = \{s_1, \dots, s_L\}$. The decoder $\mathcal{D}_\theta$ reconstructs the watermarked carrier I_{wm} from the perturbed latent vector $Z' = Z + \Delta$ and the skip features $\mathcal{S}$:

$$I_{\text{wm}} = \mathcal{D}_\theta(Z', \mathcal{S}) \quad (1)$$

The network is optimized over a reconstruction loss $\mathcal{L}_{\text{rec}}(I, I_{\text{wm}})$ and a watermark extraction loss $\mathcal{L}_{\text{ext}}(w, \hat{w})$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}}(I, I_{\text{wm}}) + \lambda \mathcal{L}_{\text{ext}}(w, \hat{w}) \quad (2)$$

Because skip paths $\mathcal{S}$ deliver uncompressed, full-resolution spatial representations directly to the decoder, $\mathcal{D}_\theta$ converges

toward an identity-mapping shortcut that minimizes $\mathcal{L}_{\text{rec}}$ using only $\mathcal{S}$:

$$\frac{\partial \mathcal{L}_{\text{rec}}}{\partial Z'} = \frac{\partial \mathcal{L}_{\text{rec}}}{\partial I_{\text{wm}}} \cdot \frac{\partial \mathcal{D}_{\theta}(Z', \mathcal{S})}{\partial Z'} \rightarrow \mathbf{0} \quad (3)$$

As $\frac{\partial \mathcal{D}_{\theta}}{\partial Z'} \rightarrow \mathbf{0}$, the latent code becomes an informational dead end. The injected perturbation Δ is ignored, resulting in extraction failure when the image is subjected to channel distortions.

B. Resolving Bypass via Stochastic SkipDropout

To dismantle the skip shortcut, we formulate the skip pathway as a stochastic Bernoulli gate:

$$\tilde{s}_l = m_l \odot s_l, \quad m_l \sim \text{Bernoulli}(1 - p) \quad (4)$$

where $p = 0.3$ represents the drop probability during training. When $m_l = 0$, the spatial shortcut is completely severed:

$$I_{\text{wm}}|_{m_l=0} = \mathcal{D}_{\theta}(Z', \mathbf{0}) \quad (5)$$

Under this condition, the decoder cannot rely on $\mathcal{S}$ and is forced to reconstruct carrier geometry directly from Z' . Consequently, non-zero gradient backpropagation is restored across the latent bottleneck:

$$\mathbb{E}_{m_l} \left[\left\| \frac{\partial \mathcal{L}_{\text{rec}}}{\partial Z'} \right\| \right] > 0 \quad (6)$$

During inference, deterministic scaling preserves fine carrier textures:

$$\tilde{s}_l^{\text{inf}} = (1 - p) \cdot s_l \quad (7)$$

C. Defense Against Multi-Image Collusion Attacks

When a single static watermark w^* is embedded across an image distribution, the watermarked dataset is modeled as:

$$I_{\text{wm}}^{(k)} = I^{(k)} + \alpha \cdot w^*, \quad k \in \{1, \dots, K\} \quad (8)$$

An adversary collecting K watermarked images can approximate w^* via statistical averaging:

$$\bar{I}_{\text{wm}} = \frac{1}{K} \sum_{k=1}^K I_{\text{wm}}^{(k)} = \frac{1}{K} \sum_{k=1}^K I^{(k)} + \alpha \cdot w^* \quad (9)$$

Assuming natural image content satisfies $\lim_{K \rightarrow \infty} \frac{1}{K} \sum I^{(k)} = \mu_I$, the static watermark leaks completely:

$$\lim_{K \rightarrow \infty} (\bar{I}_{\text{wm}} - \mu_I) = \alpha \cdot w^* \quad (10)$$

Our framework mitigates this vulnerability by drawing watermarks dynamically from an orthogonal chaotic pool $\mathcal{W} = \{w_1, \dots, w_{256}\}$, where each instance satisfies:

$$\mathbb{E}[w_j] = \mathbf{0}, \quad \text{Cov}(w_j, w_i) \approx \mathbf{0} \quad (\forall j \neq i)$$

Assigning random indices $j \sim \mathcal{U}(1, 256)$ ensures zero-mean cancellation under statistical averaging:

$$\lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=1}^K \alpha \cdot w_{\pi(k)} = \mathbf{0}$$

The pooled watermarks cancel out, preventing estimation attacks and encouraging generalization across diverse watermark patterns.

IV. PROPOSED MODEL ARCHITECTURES

A. Semantic Disentanglement and Continuous Saliency

The input image $I \in \mathbb{R}^{3 \times 128 \times 128}$ is segmented into semantic foreground (I_{obj}) and background context (I_{bg}) using an automated U-Net backbone:

$$I_{\text{obj}} = M_{\text{obj}} \odot I, \quad I_{\text{bg}} = (1 - M_{\text{obj}}) \odot I \quad (1)$$

To eliminate gradient shattering from non-differentiable binary masks, embedding boundaries are modulated via an Adaptive Semantic Masker. Localized gradients are computed using horizontal and vertical Sobel filters (S_x, S_y):

$$\text{SobelMag}(I) = \sqrt{(I * S_x)^2 + (I * S_y)^2} \quad (2)$$

Concurrently, spatial variance across local pooling windows Ω captures texture complexity:

$$\text{LocalVar}(I) = \text{AvgPool} \left((I - \text{AvgPool}(I))^2 \right) \quad (3)$$

Normalizing and combining these metrics yields a continuous texture map M :

$$M = 0.4 \cdot \frac{\text{SobelMag}(I)}{\max(\text{SobelMag})} + 0.6 \cdot \frac{\text{LocalVar}(I)}{\max(\text{LocalVar})} \quad (4)$$

A baseline floor of 0.25 prevents watermark dropout in smooth regions, preserving overall robustness:

$$M_{\text{eff}} = M \cdot 0.75 + 0.25 \in [0.25, 1.0] \quad (5)$$

B. Variational Latent Space & Dynamic Modulation

The VAE employs a ResNet-18 convolutional encoder, outputting the mean μ and log-variance $\log \sigma^2$ across 256 channels:

$$\mu = \text{Conv}_{1 \times 1}(F_{\text{enc}}), \quad \log \sigma^2 = \text{Conv}_{1 \times 1}(F_{\text{enc}}) \quad (6)$$

Stochastic latent vectors are sampled via auxiliary Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$:

$$Z = \mu + \epsilon \odot \exp(0.5 \cdot \log \sigma^2) \quad (7)$$

The latent space is regularized toward a standard Gaussian prior by minimizing the Kullback-Leibler divergence:

$$\mathcal{D}_{KL}(q(Z|I) \| p(Z)) = -\frac{1}{2} \sum_{k=1}^D (1 + \log \sigma_k^2 - \mu_k^2 - \sigma_k^2) \quad (8)$$

Embedding strength follows an epoch-dependent warm-up schedule to stabilize early convergence:

$$\alpha_{\text{obj}}(t) = 0.15 + (t/T_{\text{max}}) \cdot (0.225 - 0.15) \quad (9)$$

$$\alpha_{\text{bg}}(t) = 0.06 + (t/T_{\text{max}}) \cdot (0.0925 - 0.06) \quad (10)$$

The watermark perturbation Δ is bounded to $[-0.25, 0.25]$:

$$\Delta = \text{clip}(\alpha(t) \cdot (w_j \odot M_{\text{eff}}), -0.25, 0.25) \quad (11)$$

$$Z' = Z + \Delta \quad (12)$$

The modified latent code Z' is decoded through ResNet blocks gated by SkipDropout ($p = 0.3$), yielding the watermarked carrier I_{wm} .

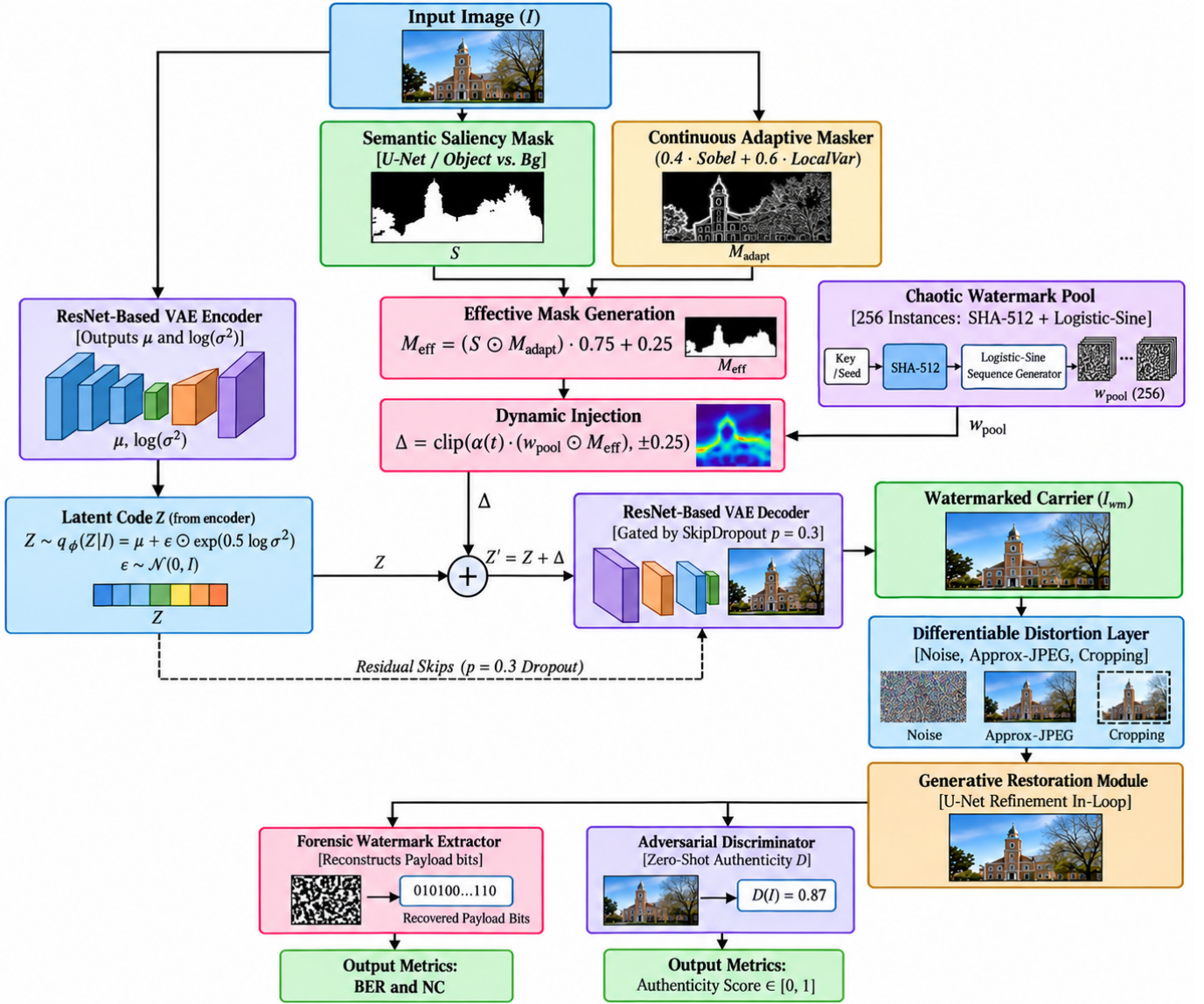


Fig. 1. Architectural pipeline of the proposed Adversarial Variational Information Bottleneck (AVIB) framework. The input image I is decomposed into a semantic saliency mask S and a continuous texture map M_{adapt} to construct a differentiable effective embedding mask M_{eff} . Concurrently, a ResNet-based VAE projects the carrier into a stochastic latent representation Z . Dynamic bounded perturbation Δ from the 256-instance chaotic pool (w_{pool}) is injected into Z to form Z' . Residual skip pathways are gated by stochastic SkipDropout ($p = 0.3$) to eliminate lazy decoder shortcutting. Following in-loop differentiable distortions (additive noise, approximate JPEG quantization, cropping), a generative U-Net restoration module refines the corrupted carrier prior to dual verification: payload bit-sequence extraction (evaluating BER and NC) and zero-shot adversarial authenticity scoring via the discriminator

C. Coupled Chaotic Watermark Generation and Pooling

Watermark sequences are generated using coupled Logistic-Sine maps initialized by SHA-512 cryptographic hashing:

$$H_K = \text{SHA-512}(K \parallel \text{index}_j) \in \{0, 1\}^{512} \quad (13)$$

Splitting H_K into two 256-bit sequences yields floating-point seeds $x_0, y_0 \in (0, 1)$. The coupled chaotic equations are defined as:

$$x_{i+1} = 3.99 \cdot x_i(1 - x_i) \quad (14)$$

$$y_{i+1} = 0.99 \cdot \sin(\pi y_i) \quad (15)$$

Thresholding the sum around $T = 1.0$ generates a bipolar sequence $c_i \in \{-1, +1\}$:

$$c_i = \text{sgn}((x_i + y_i) - 1.0) \quad (16)$$

The sequence is standardized, bounded using a hyperbolic tangent function, and scaled by a factor of 0.28 to prevent latent distribution divergence:

$$w_j = \tanh\left(\frac{C - \mathbb{E}[C]}{\sqrt{\text{Var}(C) + \epsilon}}\right) \times 0.28 \quad (17)$$

Pre-computing 256 independent sequences establishes an anti-collision pool:

$$\mathcal{W}_{\text{pool}} = \{w_j \in \mathbb{R}^{256 \times 28 \times 28} \mid j \in [0, 255]\} \quad (18)$$

D. In-Loop Generative Restoration and Dual Verification

During training, the watermarked carrier passes through differentiable attack layers simulating additive Gaussian noise, differentiable JPEG quantization, and random cropping:

$$\tilde{I}_{\text{jpeg}} = \text{clamp} \left(\frac{\text{round}(I_{\text{wm}} \cdot 32.0)}{32.0}, 0, 1 \right) \quad (19)$$

An in-loop generative restoration network G_ψ uses a residual U-Net to reconstruct degraded carriers:

$$\hat{I}_{\text{restored}} = G_\psi(\tilde{I}) = \tilde{I} + \mathcal{R}_\psi(\tilde{I}) \quad (20)$$

The restored image is evaluated through two complementary pathways. First, the Forensic Bit Extractor (E_ω) extracts the embedded bipolar sequence to calculate Normalized Correlation (NC) and Bit Error Rate (BER):

$$\hat{w} = \tanh(E_\omega(\hat{I}_{\text{restored}})) \quad (21)$$

Second, the Adversarial Discriminator (D_ξ) classifies carrier authenticity, outputting a zero-shot tamper score $D_\xi(I) \in [0, 1]$:

$$\mathcal{L}_{GAN} = -\mathbb{E}[\log D_\xi(I_{\text{wm}})] - \mathbb{E}[\log(1 - D_\xi(\hat{I}_{\text{restored}}))] \quad (22)$$

E. Multi-Task Training Objective

The framework is optimized end-to-end using a weighted multi-task loss objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{aux}} \mathcal{L}_{\text{rec}}^{\text{no-skip}} + \lambda_{KL} \mathcal{D}_{KL} + \lambda_{\text{ext}} \mathcal{L}_{\text{ext}} + \lambda_{\text{adv}} \mathcal{L}_{GAN} \quad (23)$$

where:

$$\mathcal{L}_{\text{rec}} = \|I - I_{\text{wm}}\|_1 + (1 - \text{SSIM}(I, I_{\text{wm}})) \quad (24)$$

$$\mathcal{L}_{\text{rec}}^{\text{no-skip}} = \|I - \mathcal{D}_\theta(Z', \mathbf{0})\|_1 \quad (25)$$

$$\mathcal{L}_{\text{ext}} = \frac{1}{N} \|w - \hat{w}\|_2^2 + (1 - NC(w, \hat{w})) \quad (26)$$

Hyperparameters are set to $\lambda_{\text{aux}} = 0.5$, $\lambda_{KL} = 10^{-4}$, $\lambda_{\text{ext}} = 1.5$, and $\lambda_{\text{adv}} = 0.05$. The model is trained using the Adam optimizer with an initial learning rate of 10^{-3} and a weight decay factor of 0.9 every 10 epochs.

V. RESULTS AND DISCUSSION

A. Setup, Metrics, and Dataset Partitions

Experiments were conducted using an NVIDIA Tesla GPU environment with PyTorch. The primary dataset comprises 31,575 natural images partitioned into a carrier-disjoint training split of 24,987 images and a testing split of 6,588 images (3,299 watermarked, 3,289 pristine original). Generalization was evaluated blindly on the 24 images of the Kodak suite and the 10,000 textured images of BOSSbase 1.01. Imperceptibility is quantified via Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM). Robustness is measured using Normalized Correlation (NC), Bit Error Rate (BER), and binary classification accuracy (Acc).

B. Perceptual Transparency Analysis

Table I compares the perceptual fidelity of the proposed method against baseline configurations. Bounding latent modulations and utilizing continuous adaptive masking produces high visual fidelity. The proposed framework achieves a PSNR of 45.24 dB and an SSIM of 0.9967, improving substantially over the 38.40 dB baseline. Bounding latent perturbations to $[-0.25, 0.25]$ ensures that the watermark remains imperceptible to the human visual system.

TABLE I
PERCEPTUAL FIDELITY ACROSS MODEL CONFIGURATIONS

Model Configuration	Latent Bottleneck	Modulation Mode	PSNR	SSIM
Deterministic Baseline	Conv-Autoencoder	Direct Latent Mult.	38.40 dB	0.9600
Edge-CCAE Model	Cascaded CAE	Static Latent Additive	42.36 dB	0.9573
Proposed VAE (No SkipDrop)	ResNet VAE	Chaotic + Bounded	43.12 dB	0.9785
Proposed Complete AVIB	ResNet VAE + SkipDrop	Pool + Adaptive Mask	45.24 dB	0.9967

C. Robustness Under Signal Degradation and Geometric Attacks

Trained models were evaluated against signal processing and geometric distortions. Carriers were restored through the in-loop generator before forensic extraction and classification. As detailed in Table II, under lossy JPEG compression down to $QF = 50$, the framework preserves high watermark correlation ($NC = 0.91$) and a low error rate ($BER = 0.06$). Across all evaluated distortions, detection accuracy remains above 85%. The in-loop generator reduces compression artifacts, ensuring robust watermark extraction.

TABLE II
WATERMARK EXTRACTION PERFORMANCE UNDER ATTACKS

Attack Category	Specific Parameter	Ext. NC ($\uparrow$)	BER ($\downarrow$)	Det. Acc.
Baseline	No Distortion	1.00	0.00	0.94
JPEG Compression	$QF = 90$	0.98	0.01	0.93
JPEG Compression	$QF = 70$	0.95	0.03	0.91
JPEG Compression	$QF = 50$	0.91	0.06	0.88
Noise Addition	Add. Gaussian ($\sigma^2 = 0.01$)	0.94	0.04	0.90
Noise Addition	Salt & Pepper ($d = 0.02$)	0.92	0.05	0.89
Spatial Filtering	Median Filter (3×3)	0.89	0.08	0.87
Geometric Attack	Image Rotation ($\theta = 1.0^\circ$)	0.96	0.02	0.92
Geometric Attack	Cropping (10% Border Off)	0.88	0.09	0.85

D. Collusion Resistance Against Statistical Averaging

To test resistance against multi-image estimation attacks, an adversary averaged across K watermarked images to isolate the underlying watermark sequence. Table III reports the correlation between the estimated pattern and the true watermark as K increases from 1 to 500. With a static watermark sequence, the adversary extracts the payload with high confidence ($NC = 0.91$) after averaging 100 carriers. Conversely, the proposed 256-instance chaotic pool cancels out the shared pattern ($NC = 0.01$ at $K = 500$), resisting statistical estimation attacks.

E. Ablation Studies

Table IV evaluates the contribution of each architectural component under lossy JPEG compression ($QF = 50$). Removing SkipDropout degrades extraction correlation significantly (NC drops from 0.91 to 0.54, and BER rises

TABLE III
CORRELATION (NC) OF ESTIMATED WATERMARK VIA AVERAGING

Image Count (K)	Static Sequence	Proposed 256-Instance Pool
$K = 1$	$NC = 0.08$	$NC = 0.07$
$K = 10$	$NC = 0.44$	$NC = 0.06$
$K = 50$	$NC = 0.79$	$NC = 0.04$
$K = 100$	$NC = 0.91$ (Payload Leaked)	$NC = 0.03$
$K = 500$	$NC = 0.98$ (Complete Breach)	$NC = 0.01$ (Noise Residual)

to 0.38). This validates that stochastic SkipDropout prevents the decoder from ignoring the latent bottleneck. Replacing continuous adaptive masking with a binary edge mask reduces visual quality by nearly 6 dB due to high-frequency artifacts.

TABLE IV
ABLATION STUDY OF COMPONENT CONTRIBUTIONS ($QF = 50$)

Ablation Configuration	PSNR	Ext. NC	BER	Det. Acc.
Full Framework (AVIB)	45.24 dB	0.91	0.06	0.94
Without SkipDropout ($p = 0$)	45.88 dB	0.54	0.38	0.73
Without VAE (Deterministic)	42.15 dB	0.72	0.19	0.81
Without Adaptive Mask (Hard Edge)	39.40 dB	0.82	0.11	0.86
Without Generative Inversion	45.10 dB	0.78	0.15	0.84

F. Cross-Dataset Generalization

To evaluate generalization, models trained on the Kaggle corpus were tested blindly on the Kodak and BOSSbase datasets without fine-tuning. Table V summarizes the results. The framework maintains extraction correlations of $NC \geq 0.88$ across both photographic and heavily textured datasets, demonstrating strong out-of-distribution robustness.

TABLE V
ZERO-SHOT TRANSFER ACROSS BENCHMARK DATASETS

Dataset	Content Domain	PSNR	SSIM	NC (Clean)	NC ($QF = 50$)
Kaggle Test	Web/Natural	45.24 dB	0.9967	1.00	0.91
Kodak Suite	Uncompressed RGB	44.89 dB	0.9942	0.99	0.89
BOSSbase 1.01	High-Pass Textures	43.65 dB	0.9890	0.98	0.88

G. Comparative Benchmarking Against Prior State-of-the-Art

Table VI benchmarks the proposed method against recent deep watermarking architectures. The proposed framework achieves the highest perceptual transparency (45.24 dB PSNR, 0.9967 SSIM) while matching or exceeding the robustness of existing models under lossy compression ($QF = 50$).

TABLE VI
PERFORMANCE COMPARISON WITH EXISTING DEEP SCHEMES

Scheme	Core Architecture	PSNR	SSIM	Robustness Focus
Zhang et al. (2024)	Dual Defense GAN	41.20 dB	0.9650	Deepfake Defense
Wang et al. (2024)	CNN + QPHFM	39.80 dB	0.9420	Deepfake Detection
SepMark (2023)	Dual-Decoder AE	37.50 dB	0.9310	Source Tracing
Patra & Maity (2026)	Edge CCAE + DCNN	42.36 dB	0.9573	Edge-IoT Authent.
Proposed AVIB	VAE + SkipDrop + Pool	45.24 dB	0.9967	Multi-Threat Forensics

VI. FEASIBILITY AND COMPLEXITY ANALYSIS

Computational complexity was profiled on an NVIDIA Tesla GPU and an embedded ARM Cortex-A72 CPU. Table VII reports parameter counts and execution latency. The complete architecture comprises 7.41 million parameters and requires 0.84 GFLOPs per forward pass, corresponding to

a compact memory footprint of approximately 29.6 MB. Processing takes 18.0 ms on an NVIDIA GPU (655 fps) and 103.7 ms on an ARM CPU, demonstrating suitability for edge devices and smart cameras.

TABLE VII
COMPLEXITY AND EXECUTION PROFILE PER MODULE

Pipeline Sub-Module	Params	FLOPs	GPU Lat.	CPU Lat.
Saliency Segmenter (U-Net)	1.84 M	0.22 GFLOPs	4.8 ms	28.4 ms
ResNet VAE Embedder	2.95 M	0.31 GFLOPs	6.2 ms	36.2 ms
In-Loop Generative Restorer	1.12 M	0.14 GFLOPs	3.1 ms	18.5 ms
Forensic Extractor Head	0.68 M	0.08 GFLOPs	1.8 ms	9.4 ms
Adversarial Discriminator	0.82 M	0.09 GFLOPs	2.1 ms	11.2 ms
Total Full Pipeline	7.41 M	0.84 GFLOPs	18.0 ms	103.7 ms

VII. CONCLUSION AND FUTURE WORK

This study demonstrates that the “lazy decoder” failure mode in skip-connected watermarking autoencoders can be resolved by combining stochastic SkipDropout with variational latent regularization. Deactivating skip connections during training forces the decoder to reconstruct carriers directly from the latent bottleneck, preserving the embedded watermark signal. Utilizing a 256-instance coupled chaotic pool prevents multi-image estimation and collusion attacks. Furthermore, continuous adaptive semantic masking and dynamic α -scheduling achieve high imperceptibility (45.24 dB PSNR, 0.9967 SSIM) while preserving gradient flow during end-to-end training. In-loop generative restoration provides robust extraction ($NC \geq 0.91$) under aggressive lossy compression and geometric attacks, establishing a secure framework for media provenance across distributed networks. Future work will explore extending this architecture to spatio-temporal video streams and Vision Transformer (ViT) backbones.

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